MILITARY SPECIFICATION

SWIVEL, BALL-BEARING (SPECIAL, WATERTIGHT)

1. SCOPE

1.1 Scope.- This specification covers a special watertight ball-bearing swivel for minesweeping applications.

2. APPLICABLE DOCUMENTS

2.1 Specifications, standards, drawings and publications.- Not applicable.

(Copies of specifications, standards, drawings and publications required by the contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications.- The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

AMERICAN IRON AND STEEL INSTITUTE
Steel Products Manual, Section 10, Alloy Steels.

(Application for copies should be addressed to the American Iron and Steel Institute, 150 East 42nd Street, New York 17, New York.)

3. REQUIREMENTS

3.1 Preproduction sample.- Prior to beginning production a sample shall be tested as specified in 4.2 (see 6.2).

3.2 General requirements.- The swivel shall have a watertight enclosure for the ball bearings and shall turn freely under load.

3.3 Design and construction.-

3.3.1 Fabrication.- All highly stressed swivel parts shall be fabricated from 4135 steel conforming to the alloy steels section of the American Iron and Steel Institute Steel Products Manual.

3.3.2 Size.- The swivel dimensions shall be in general in accordance with figure 1. Principal dimensions only have been indicated in order that the same dies and tooling used in the manufacture of similar commercial design swivels can be utilized.

3.3.3 Finish.- All external surfaces shall be cadmium plated to prevent corrosion. All sharp edges and burrs shall be removed prior to plating.

3.3.4 Bearing packing.- The encased ball bearings shall be packed with a heavy lifetime grease at assembly. No voids or air pockets shall remain in the bearing enclosure.

3.3.5 Seals.- The bearing enclosure shall be watertight to withstand a pressure of 180 pounds per square inch (psi).

3.3.6 Lock.- After assembly the bearing enclosure and eye shall be locked by an socket drive set screw.

3.3.7 Operation.- The swivel shall be capable of supporting a 10,000 pound static load and shall swivel freely under a 5,000 pound load.

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3.3.8 Identification.- The following data shall be stamped on the bearing enclosure in 1/8-inch letters:

Stock No._____
Contract No._____
Manufacturer's Name or Trademark_____
Working Load: 2-1/2 tons
Proof load: 5 tons

4. QUALITY ASSURANCE PROVISIONS

4.1 Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. The government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Preproduction inspection.- Preproduction inspection shall consist of the examination and tests specified in 4.4.1 through 4.4.3.

4.3 Quality conformance inspection.-

4.3.1 Lot.- For purposes of sampling, a lot shall consist of all swivels offered for delivery at one time.

4.3.2 Sampling for quality conformance inspection.- From each lot a random sample of swivels shall be selected as shown in table I and subjected to the tests specified in 4.4.1 and 4.4.2, respectively.

Table I - Sampling for quality conformance inspection.

Number of swivels in inspection lot	Number of swivels in sample
1-8	1
9-15	2
16-30	3
31-65	5
66-110	7
111-180	10
181-300	15
301-500	25

4.4 Test procedures.-

4.4.1 Proof test.- The swivel shall be subjected to a static tensile load of 10,000 pounds. Fracture or permanent deformation of any part of the swivel shall be cause for rejection.

4.4.2 Working load test.- The swivel shall be tested to determine its ability to turn freely by hand, using a bar 12-inches long, under a tensile load of 5,000 pounds. Failure of the swivel to conform to this test shall be cause for rejection.

4.4.3 Watertightness test.- The swivel shall be submerged in water for two hours at a pressure of 180 psi. The swivel shall then be disassembled to determine whether any leakage has occurred. Presence of any water in the swivel interior shall be cause for rejection.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, packing, and marking.- Preservation, packaging, packing, and marking shall be as specified in the contract or order (see 6.1).

6. NOTES

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6.1 Ordering data.- Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Preservation, packaging, packing and marking required.

6.2 Preproduction.-

6.2.1 Invitations for bids should provide that the Government reserves the right to waive the requirement for preproduction samples as to those bidders offering a product which has been previously procured or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending procurement.

Notice.- When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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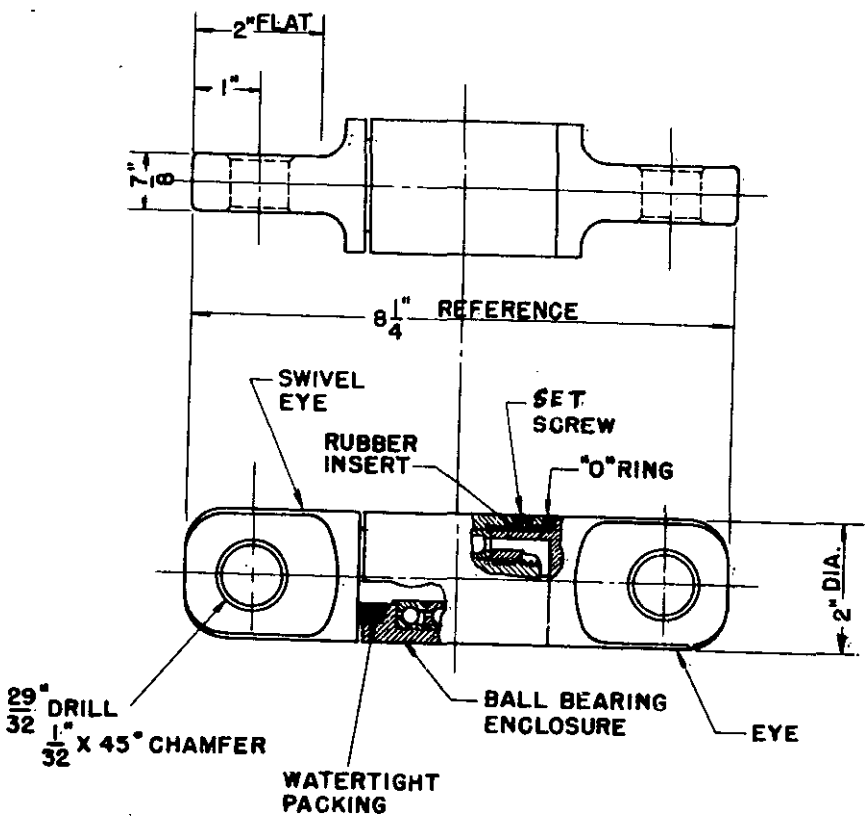


Figure 1 - Ball bearing swivel.

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